

GUSTAVO VARELA ALVARENGA, Ph.D.

Data Scientist & Statistician

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SUMMARY

Data scientist and statistician (Ph.D.) with 20+ years across industry, research, and academia, focused on turning messy data into decisions. Most recently a Senior Data Science Researcher at Google; previously founded and led the Latin American data science team at Argus Media. Deep expertise in time-series forecasting, machine learning, predictive modeling, and the design and deployment of data products in R, Python, and SQL, with a track record of translating complex analytics for non-technical stakeholders.

CORE SKILLS

Programming & Tools: R, Python, SQL, SAS, R Shiny, Git/GitHub, AWS, PostgreSQL, MySQL

Methods: Time-Series Forecasting, Machine Learning, Predictive Modeling, Deep Learning (CNNs/RNNs), Econometrics, Causal Inference (Difference-in-Differences, Propensity Score Matching), Regression & ANOVA, Monte Carlo Simulation, Data Visualization

Leadership: Team Building & Management, Stakeholder Engagement, Data Product Development, Quality Assurance

Languages: English (fluent), Portuguese (native)

EXPERIENCE

Senior Data Science Researcher — Google

Jun 2025 – Jun 2026

- Designed a novel metric for evaluating content on the Search results page, enabling systematic assessment of content quality and pinpointing where relevant content was missing.
- Assessed where Google's AI Mode could benefit from AI agents, defining the cases in which agentic information retrieval would better fulfill user queries.
- Evaluated the landscape of grounding and corroboration sources for AI Overviews, recommending higher-quality sources to validate AI-generated information.

Regional Head of Data Science for Latin America — Argus Media

Jan 2021 – Feb 2025 | São Paulo, Brazil

- Founded and led the Latin American data science team, growing it to 12 data engineers and scientists across three departments, with a department head and team lead reporting directly.
- Drove an 86% team-retention rate over nearly four years by building a culture of innovation and collaboration.
- Spearheaded the creation and deployment of R-based data products and scalable R/Shiny applications on AWS, improving client decision-making and operational efficiency through automation.
- Owned daily production of the two principal data science products, ensuring model integrity and accuracy of results prior to publication.
- Instituted quality-assurance protocols and strict Git version control for the top revenue-generating product, improving reliability and performance.
- Partnered with business development and sales to design and present data science solutions to clients, increasing engagement and product adoption.

Lecturer — University of Maryland

Jan 2020 – May 2021

- Developed and taught a 400-level Computational Statistics course covering data wrangling, exploratory data analysis, and inferential modeling using SAS and SQL.
- Designed data-backed research projects and evaluation strategies, guiding students through regression, logistic regression, and ANOVA.

NSF Mathematical Sciences Graduate Intern — Pacific Northwest National Laboratory (PNNL)

Jun 2019 – Aug 2019

- Proposed and prototyped a PostgreSQL/MySQL RDBMS for a DOE/NETL/PNNL project, optimizing database architecture for processing efficiency.
- Engineered a Python web crawler and a web application for experimental data input, improving data-collection accuracy.
- Led an R programming workshop (blogdown, Shiny) for an all-women audience.

Lead Statistician — Agência Brasileira de Desenvolvimento Industrial (ABDI)

Aug 2013 – Jul 2014

- Evaluated the impact of public policies on Brazilian industrial development using simulation and quasi-experimental methods (difference-in-differences, propensity score matching), presenting findings to the Brazilian Minister of Industry.
- Led collection and analysis of socioeconomic data informing policy recommendations; contributed to the web tool “The Competitiveness Decoder.”

Lecturer — Universidade de Brasília

Aug 2012 – Jul 2014

- Taught statistical modeling to undergraduates from non-STEM backgrounds, improving applied data-analysis skills through interdisciplinary, project-based methods.

Statistical Consultant — IPEA (Institute for Applied Economic Research)

Jan 2009 – Aug 2013

- Designed and executed studies using advanced statistical and econometric methods to evaluate Brazilian industrial policies, producing strategic recommendations.
- Built data extraction and management pipelines, including web scraping for patent data and phonetic matching for dataset integration.
- Advised senior researchers on statistical methodology across the research lifecycle.

EDUCATION

Ph.D., Mathematical Statistics — University of Maryland | 2014 – 2021

Dissertation: Innovations in Time Series Forecasting — New Validation Procedures to Improve Forecasting Accuracy and a Novel Machine-Learning Strategy for Model Selection.

M.A., Economics — Universidade de Brasília | 2009 – 2011

B.S., Statistics — Universidade de Brasília | 2004 – 2008

HONORS & CERTIFICATIONS

Herbert A. Hauptman Fellowship, University of Maryland (2021)

Prêmio CNI de Economia (CNI National Economy Award), 2nd place (2012)

Certifications: Neural Networks & Deep Learning (Coursera), Game Theory (Stanford/Coursera), Model Thinking, Project Management with Monday.com